Digital Egypt Pioneers Initiative (DEPI)

Graduation Project | First Deliverable

CareLens

Healthcare Service Utilization, Risk, and Cost Analytics

Team: skills dynamix

Group ID: ALX5_DAT3_S3

Track: Data Analytics – Professional Data Analyst

1. Team Information

#	Member Name (as registered)	Track
1	حمزة أسامة محمد محمد	Data Analytics – Professional Data Analyst
2	سيف الدين خالد احمد منير	Data Analytics – Professional Data Analyst
3	تقي الدين حسين احمد السيد	Data Analytics – Professional Data Analyst
4	احمد محمود محمد محمد	Data Analytics – Professional Data Analyst
5	محمد احمد راشد عطية	Data Analytics – Professional Data Analyst
6	محمد محمود عبد القادر	Data Analytics – Professional Data Analyst

Team name / company: skills dynamix. Group ID: ALX5_DAT3_S3.

2. Project Summary

Project title	CareLens: Healthcare Service Utilization, Risk, and Cost Analytics
Domain	Healthcare Analytics
Data source	Hospital Inpatient Discharges (SPARCS De-Identified), 2022 and 2023, State of New York health open data portal

3. Project Overview

This project analyzes healthcare service utilization data to understand patient demand patterns and to evaluate the performance of the healthcare system. It examines temporal, geographic, and condition-related patterns that are associated with higher risk factors and higher service costs, and aims to translate the findings into data-driven operational insights for healthcare resource management.

4. Problem Statement

Healthcare providers and administrators must plan capacity, staffing, and budgets while demand for services varies across time, locations, and patient conditions. Without a structured analysis of utilization data, it is difficult to identify when and where demand concentrates, which conditions are linked to elevated risk and cost, and where system performance can be improved.

This project addresses that gap by analyzing available utilization data to surface these patterns and make them accessible to decision-makers.

5. Objectives

5.1 Main Objective

Analyze healthcare service utilization data to understand patient demand patterns and evaluate healthcare system performance, in order to provide data-driven operational insights for healthcare resource management.

5.2 Specific Objectives

- Identify temporal patterns in service utilization (for example, daily, weekly, or seasonal variation).
- Identify geographic differences in demand and performance.
- Analyze condition-related patterns and their association with higher risk factors.
- Examine the factors associated with higher service costs.
- Present the findings in a form that supports operational and resource-planning decisions.

6. Key Analytical Questions

1. How does patient demand for healthcare services vary over time?
2. How does utilization and system performance vary across geographic areas?
3. Which conditions are associated with higher risk factors and higher service costs?
4. Which operational indicators best describe healthcare system performance?
5. What actionable insights can support healthcare resource management?

7. Target Users and Stakeholders

- Healthcare administrators and operations managers responsible for resource and capacity planning.
- Healthcare policy and planning decision-makers.
- Clinical and operational teams who use utilization insights to plan services.

8. Data

The project uses the Hospital Inpatient Discharges (SPARCS De-Identified) datasets for 2022 and 2023, published on the State of New York health open data portal (health.data.ny.gov).

Year	Dataset	Link
2022	Hospital Inpatient Discharges (SPARCS De-Identified): 2022	https://health.data.ny.gov/Health/Hospital-Inpatient-Discharges-SPARCS-De-Identified/5dtw-tffi/about_data
2023	Hospital Inpatient Discharges (SPARCS De-Identified): 2023	https://health.data.ny.gov/Health/Hospital-Inpatient-Discharges-SPARCS-De-Identified/46xm-urtu/about_data

Data quality, missing values, and potential limitations of the datasets will be documented after the initial data inspection.

9. Proposed Methodology and Workflow

Step	Phase	Description
1	Data Collection	Obtain the 2022 and 2023 SPARCS datasets from the official portal and document their structure and source.
2	Data Cleaning & Preprocessing	Handle missing values, duplicates, inconsistent formats, and outliers.
3	Exploratory Data Analysis	Examine distributions and relationships; analyze temporal, geographic, and condition-related patterns.
4	Risk and Cost Analysis	Investigate factors associated with higher risk and service costs.
5	Visualization / Dashboard	Present findings in a dashboard or report.
6	Insights & Recommendations	Summarize operational insights for healthcare resource management.

10. Expected Impact

The project is expected to give healthcare decision-makers a clearer, evidence-based view of demand patterns, high-risk and high-cost areas, and performance gaps, supporting more informed planning of healthcare resources.